Vision Based Navigation

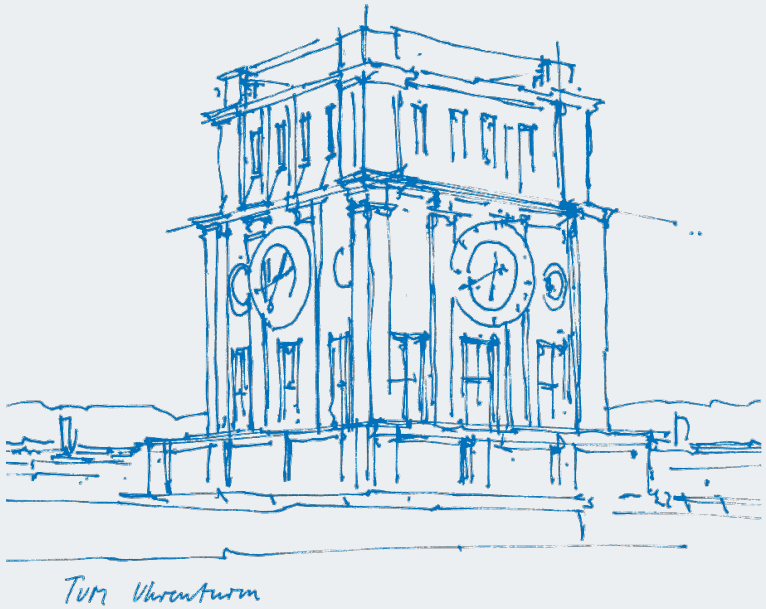
Computer Vision Group - TUM

16.11.2026

Practical Course: Vision Based Navigation

Project Phase

Mateo de Mayo, Jonathan Eichhorn
Prof. Dr. Daniel Cremers



Date: 16.11.2026

Project Phase

Projects:

- Start after sheet 5 is complete. For the remainder of the lecture period.
- Work on open-ended project (1-2 people).
- Registrate your group and select your project topic here by 30.11.2026
- We will assign your group a tutor before 07.12.2026 (day of first session)
- Mandatory weekly meeting with tutors to discuss progress and next steps.
  - Fixed 30 min time slot
  - Mondays between 14:00-18:00
- Project goal is to be determined:
  - Choose from list of suggested projects or suggest your own.
  - "Advanced" topics have more uncertain scope / solution. More independent work required.
  - At most 2 groups should work on the same project (first come first served).

Presentation:

- Present project outcome in talk and Q&A session (15min + 5min)
- Written report on project outcome (10-12 pages, single column, single-spaced lines, 11 pt)

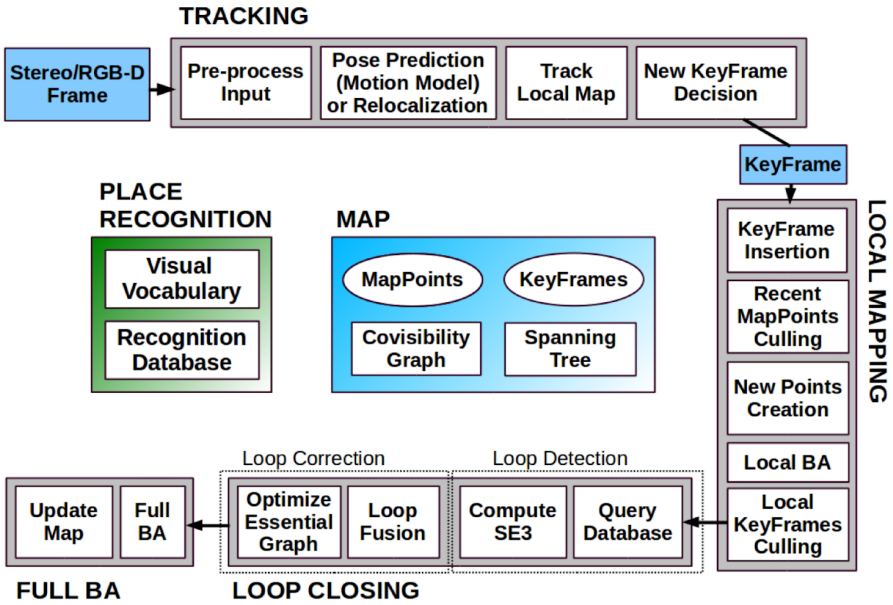
Important dates:

- Fix groups, project topic, and time for weekly meeting: 30.11.2026
- First meeting on 07.12.2026
- Other meeting dates on 14.12, 21.12, 11.01, 18.01, 25.01
- Project presentations: 01.02.2027 14:00
- Project report due: 08.02.2027

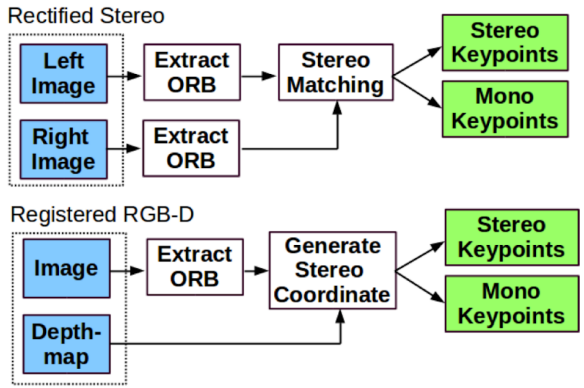
Registration Spreadsheet

- Go to spreadsheet: https://docs.google.com/spreadsheets/d/1lMjKCFbPl6oY4u12pocHEJFguWC3f1i80bqL_08at_c/edit

1. Loop closure



(a) System Threads and Modules.

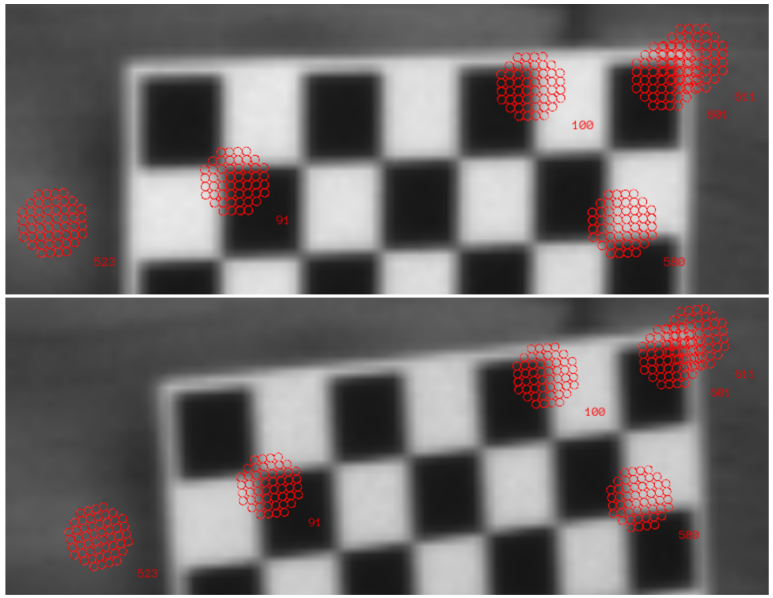


(b) Input pre-processing

- ORB_SLAM: http://webdiis.unizar.es/~raulmur/MurMontielTardosTRO15.pdf
- ORB_SLAM2: https://arxiv.org/abs/1610.06475
- Map management
- Reusing Keyframes
- Spanning tree for pose-graph optimization

2. Indirect Visual Odometry with Optical Flow

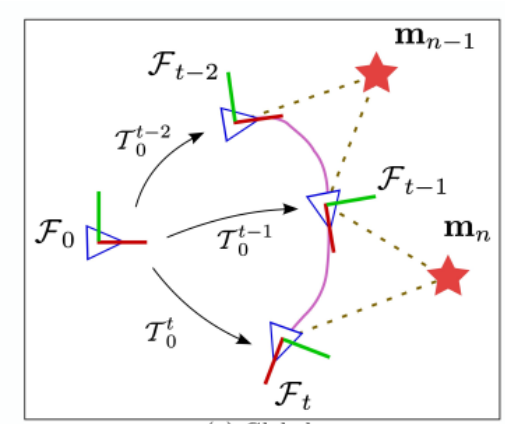
- Sparse optical flow as alternative to feature matching
- Extend odometry application
- Compare runtime, accuracy, ...
- Possible extensions:
  - patch similarity norms
  - Keyframing, local optimization
  - Different image warping strategies
  - Implement Gauss-Newton (or LM) manually



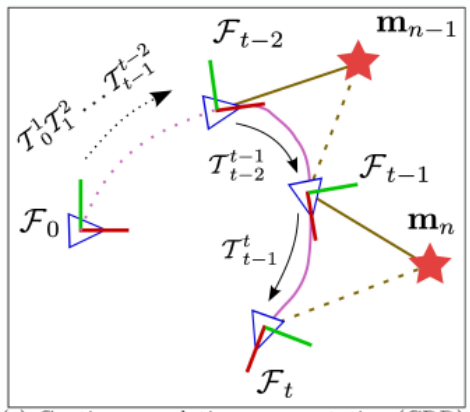
- Visual-Inertial Mapping with Non-Linear Factor Recovery (V. Usenko, N. Demmel, D. Schubert, J. Stueckler and D. Cremers), In arXiv:1904.06504, 2019. https://arxiv.org/pdf/1904.06504
- Equivalence and efficiency of image alignment algorithms (Baker, Simon, and Iain Matthews), In IEEE Computer Society Conference on Computer Vision and Pattern Recognition. Vol. 1. IEEE Computer Society; 1999, 2001. http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.70.20&rep=rep1&type=pdf
- Pyramidal Implementation of the Lucas Kanade Feature Tracker, Description of the algorithm (Jean-Yves Bouguet), Intel Corporation. https://robots.stanford.edu/cs223b04/algo_tracking.pdf

3. Relative Map Formulation for SLAM

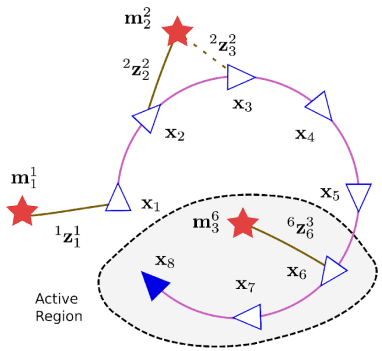
- Change the map formulation to the relative one
  - Parameters are relative poses between keyframes
  - All points are defined relative to some frame
- Extend either SfM or Odometry application
- Paper: http://www.robots.ox.ac.uk/~mobile/Papers/2010IJCV_mei.pdf



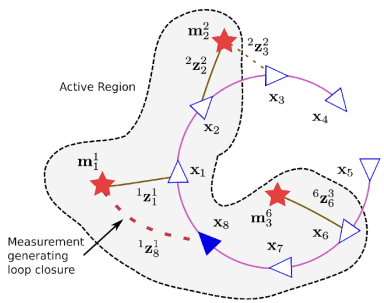
(a) Global



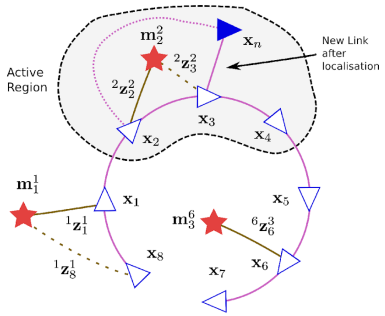
(e) Continuous relative representation (CRR)



(a)



(b)



(c)

4. Advanced Matching and Keypoint Evaluation

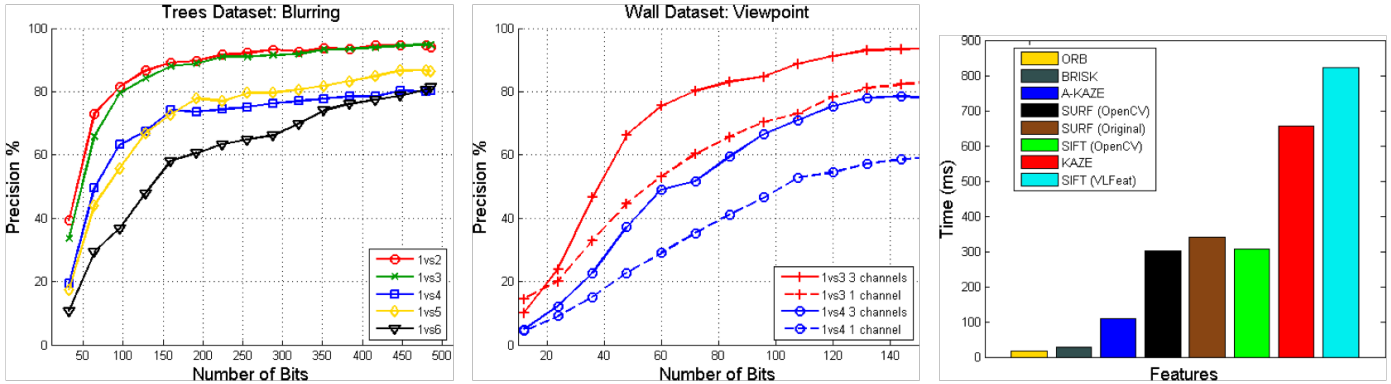
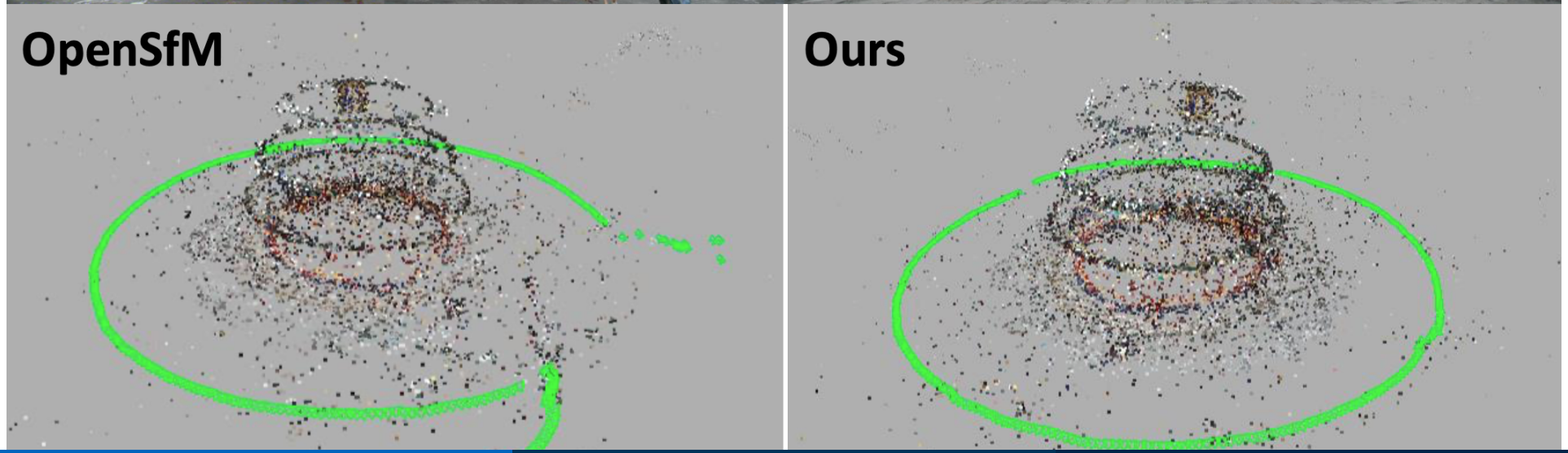


Figure 1: Alcantarilla, Pablo F., and T. Solutions. "Fast explicit diffusion for accelerated features in nonlinear scale spaces." IEEE Trans. Patt. Anal. Mach. Intell 34.7 (2011): 1281-1298.

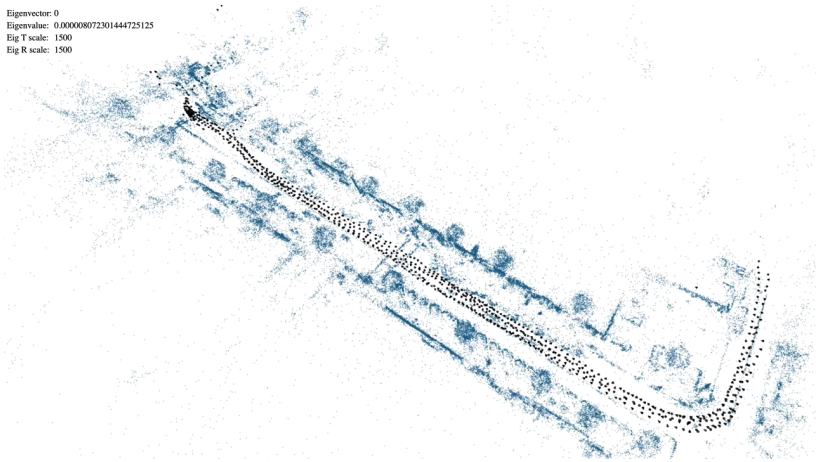
- Keypoints evaluation:
  - ORB, AKAZE, SIFT, BRISK
  - Computation time / matching statistics
- Cascade Hashing for descriptor matching:
  - http://www.nlpr.ia.ac.cn/jcheng/papers/CameraReady-CasHash.pdf

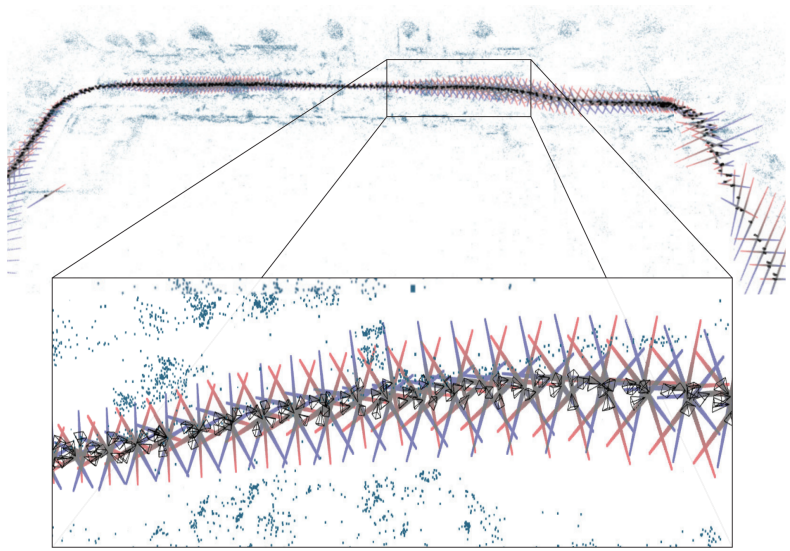
5. Improving SfM with Reliable Resectioning (**advanced**)





6. Visualizing Spectral BA Uncertainty (**advanced**)

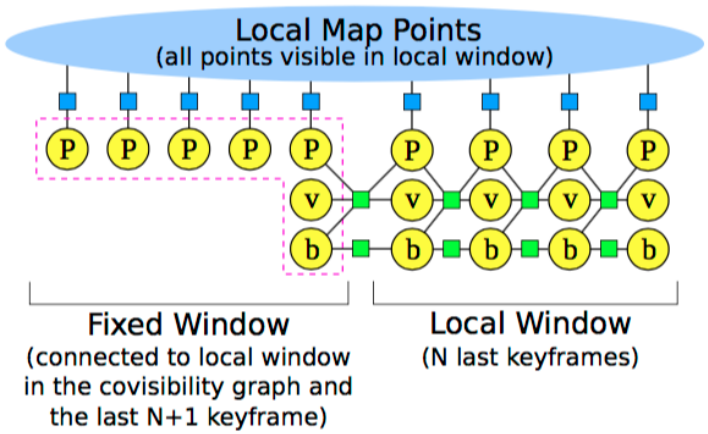


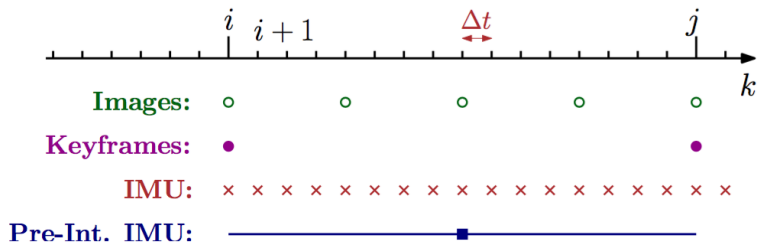


- K. Wilson and S. Wehrwein, 3DV 2020
  - https://facultyweb.cs.wwu.edu/~wehrwes/files/sfmflex.pdf
  - video: https://slideslive.com/embed/presentation/38941084
- Implement pose uncertainty visualisation in our SfM application
  - static (ellipses)
  - animated
- Investigate relative scaling of units (rotation vs translation)
- public implementation for reference: https://wilsonkl.github.io/sfmflex-release/

7. Visual-Inertial Tracking using Preintegrated Factors (**advanced**)

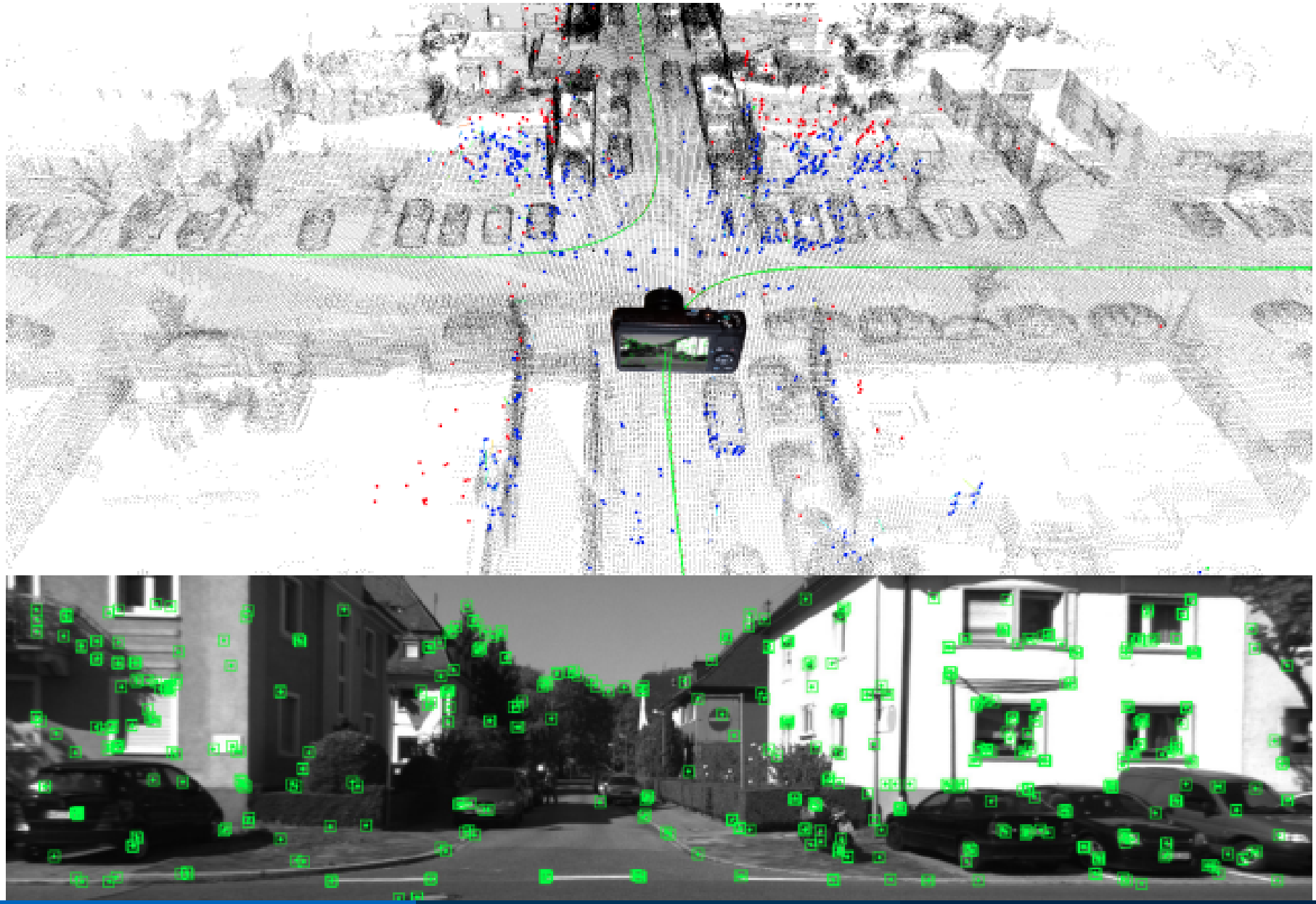
- Use camera + IMU for stability and scale observability
- Estimate IMU biases and velocity
- Preintegrate measurements between image frames



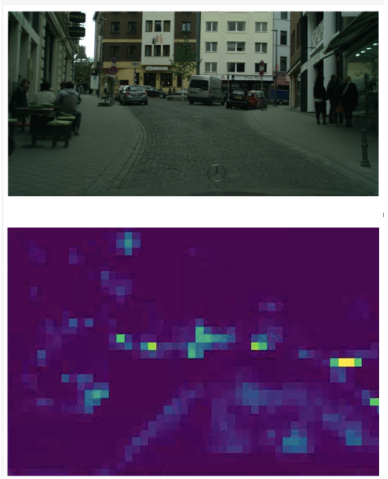


- Theory: Forster et al., "On-manifold preintegration for real-time visual-inertial odometry", 2016 http://rpg.ifi.uzh.ch/docs/TRO16_forster.pdf
- Library with preintegrated factors: <gtsam.org>
- Example system with preintegrated factors: visual-inertial ORB-SLAM https://arxiv.org/pdf/1610.05949.pdf

8. Monocular Camera Localization in 3D LiDAR Maps



9. Evaluating feature uncertainty representations in the covariance matrix



- There are different sources of uncertainty coming from the observations Photometric uncertainty:
  - descriptor distance / patch intensity difference
  - Model uncertainty: reprojection and epipolar errors
  - Outlier detection: ransac, ignoring quartiles
- This project looks for the best and fastest ways of quantifying the uncertainties Requires implementing the weight matrix, modifying it in different ways (to be discussed with the instructors), and making a thorough evaluation of the results
- PS: if you are looking for a potential thesis, you would like to come up with experiment ideas and evaluating them

10. Hand crafted optimization (**advanced**)

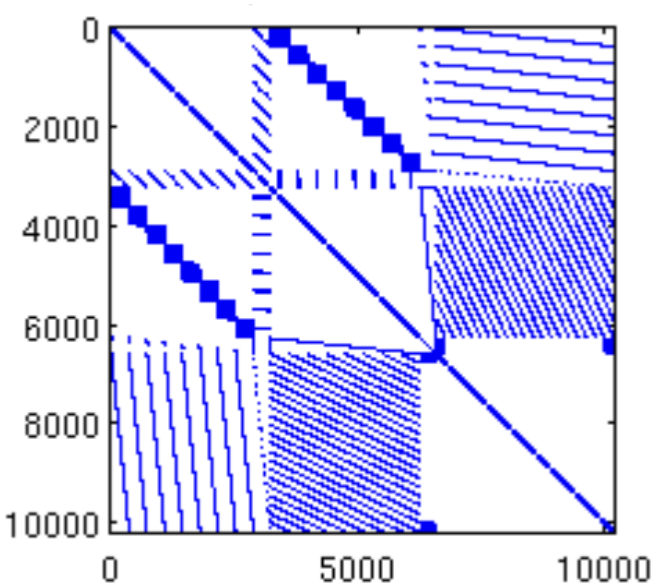


Figure 3: Hessian of reprojection loss

$$\begin{aligned}\frac{\partial (\mathbf{Rp})}{\partial \boldsymbol{\varphi}} &= \lim_{\boldsymbol{\varphi}\to 0} \frac{\exp\left(\boldsymbol{\varphi}^{\wedge}\right)\exp\left(\boldsymbol{\phi}^{\wedge}\right)\mathbf{p} - \exp\left(\boldsymbol{\phi}^{\wedge}\right)\mathbf{p}}{\boldsymbol{\varphi}} \\ &= \lim_{\boldsymbol{\varphi}\to 0} \frac{\left(\mathbf{I}+\boldsymbol{\varphi}^{\wedge}\right)\exp\left(\boldsymbol{\phi}^{\wedge}\right)\mathbf{p} - \exp\left(\boldsymbol{\phi}^{\wedge}\right)\mathbf{p}}{\boldsymbol{\varphi}} \\ &= \lim_{\boldsymbol{\varphi}\to 0} \frac{\boldsymbol{\varphi}^{\wedge}\mathbf{Rp}}{\boldsymbol{\varphi}} = \lim_{\boldsymbol{\varphi}\to 0} \frac{-(\mathbf{Rp})^{\wedge}\boldsymbol{\varphi}}{\boldsymbol{\varphi}} = -(\mathbf{Rp})^{\wedge}\end{aligned}$$

Figure 4: Derivation of rotated 3D point w.r.t rotation

- Remove the dependency on Ceres and make things fast
- You will need to derive all jacobians in the process And you will need to make a data-oriented implementation that leverages the sparsity of the problem
- PS: you like making things go fast and using c++ for performance. You like the idea of implementing numerical optimization methods and deriving jacobians.

11. Online multi-sensor temporal alignment

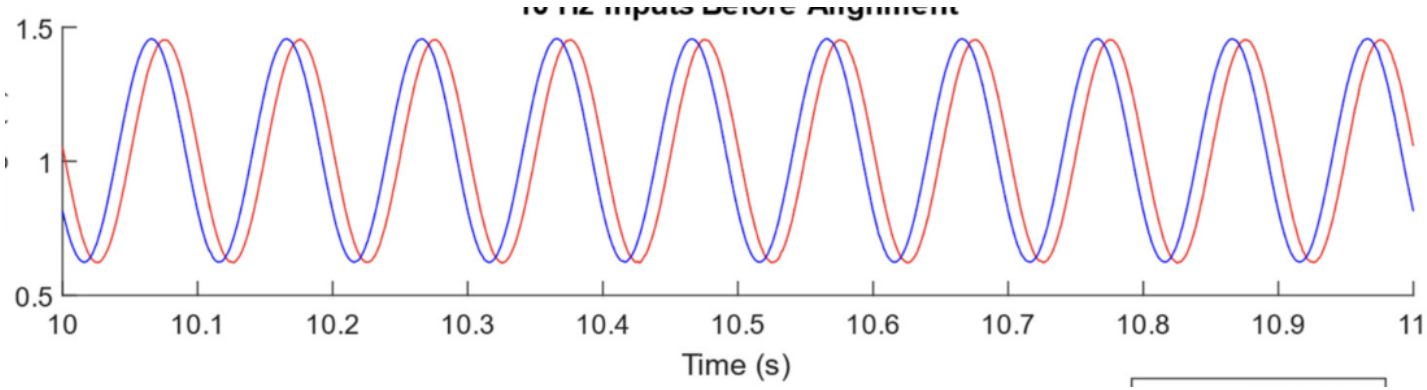


Figure 5: Red and blue curve are representing two measurements from two different sensor

- Initial focus on aligning IMU and camera unsynced timestamps (similar scenario to e.g. phones) This is NOT visual-inertial odometry, just producing an temporal offset for each frame, and minimizing the error against the known offset Expandability to aligning other trajectories like mocap-camera, gps-camera
- Given datasets with both aligned and unaligned timestamps, from the unaligned, in realtime get the temporal alignment offset.
- Should be fast and isolated How could it be done? IMU-camera aligning rotational velocities, gps-camera: aligning trajectories How can it be improved? study answers to: when to trigger the alignment to minimize performance hit? how many samples to use? moving window? etc
- PS: you like to derive and implement a simple optimization of your own, you would like to tackle a real problem and maybe contribute to an open source system

Other Projects

Other possibilities, **we do not guarantee that we will allow them**:

Recommend Your Own Project

- If you have an idea for a project that is not on the list, feel free to discuss it with the instructors.
- We are open to any project that is related to vision-based navigation and is feasible within the given time frame.
- If you have a specific paper or topic in mind, please share it with us and we can discuss how to implement it as a project.

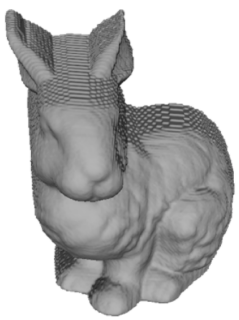
Combine Projects with Thesis or Guided Research

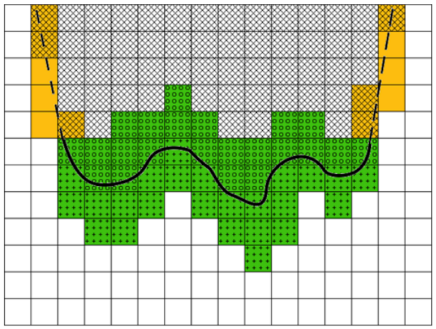
- If you think you might want to do a thesis or guided research with any of us the lecturers, you can use the project phase to explore a potential topic and get a head start on your research.
- However, keep in mind that the project phase is meant to be a learning experience and is not a guarantee of a thesis topic or research position. You should still be open to exploring other topics and working with other lecturers if you find something that interests you.

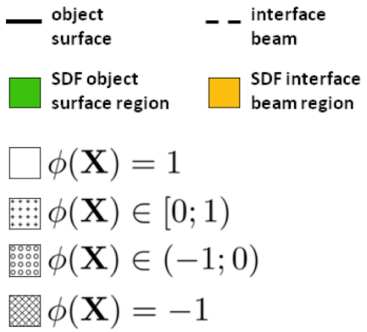
Deprecated Projects

These projects are not recommended, but if you are interested in them, feel free to discuss with the instructors.

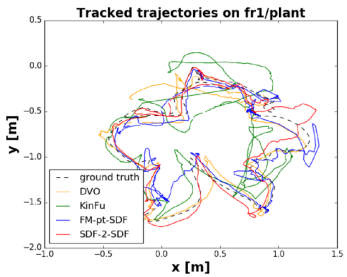
SDF-based Tracking and Reconstruction (**advanced**)

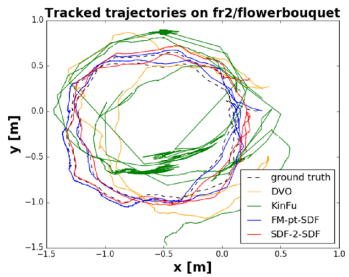


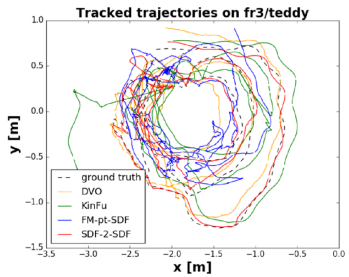




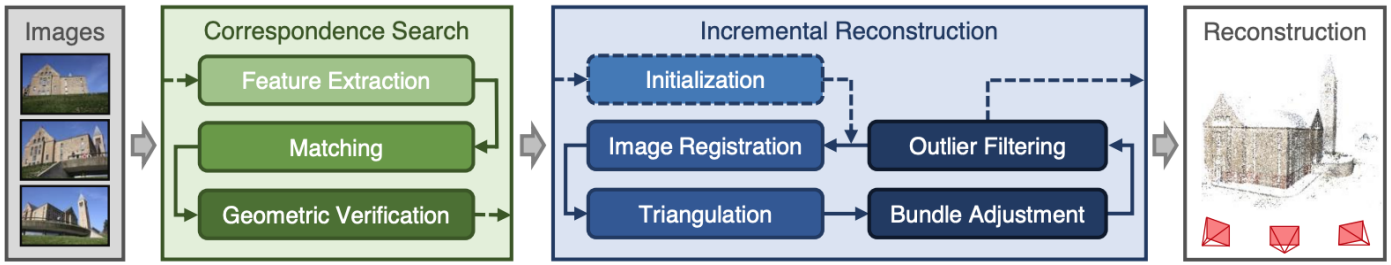
- Signed distance function (SDF) for surface representation
- Implement SDF-based tracking in reconstruction pipeline
- Slavcheva et al., "Sdf-2-sdf: Highly accurate 3d object reconstruction", ECCV 2016 http://campar.in.tum.de/pub/slavcheva2016eccv/slavcheva2016eccv.pdf
- Bylow et al., Real-time camera tracking and 3D reconstruction using signed distance functions, RSS 2013 https://vision.cs.tum.edu/_media/spezial/bib/bylow_etal_rss2013.pdf





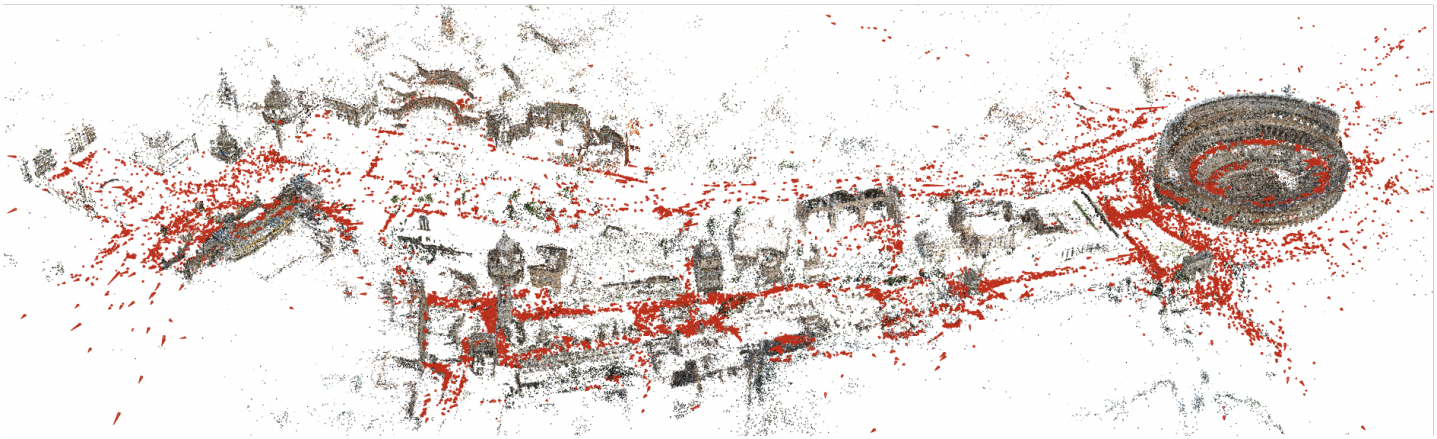


Structure from Motion Revisited



- Improve different stages of SfM application
  - Initialisation
  - Next-best-view selection
  - Triangulation
  - Re-triangulation and Outlier Filtering
- Schonberger, Johannes L., and Jan-Michael Frahm. "Structure-from-motion revisited." Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition. 2016. (https://www.cv-foundation.org/openaccess/content_cvpr_2016/papers/Schonberger_Structure-From-Motion_Revisited_CVPR_2016_paper.pdf)

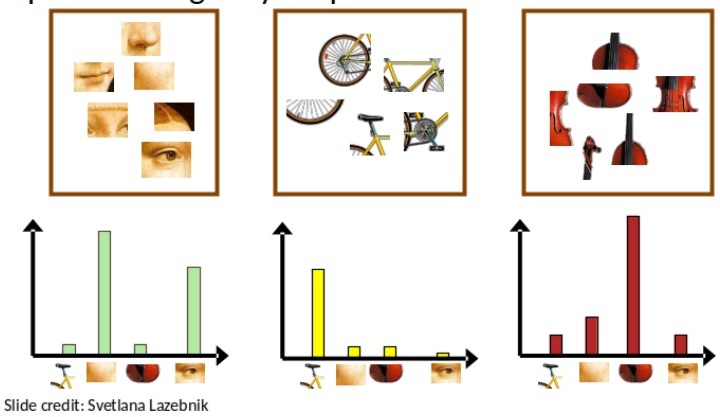
SfM for Large Image Collections (**advanced**)



- Make SfM work for thousands of images
  - Full Euroc sequences or other datasets
  - Select subset of images
    - Subsample
    - Discard too close or similar images
  - Use BoW for candidate selection
  - Implement direct index for more efficient feature matching
  - More efficient geometric verification
    - Schönberger J.L., Price T., Sattler T., Frahm JM., Pollefeys M. A Vote-and-Verify Strategy for Fast Spatial Verification in Image Retrieval. ACCV 2016. (https://demuc.de/papers/schoenberger2016vote.pdf)

Bag of Words for Place Recognition

1. Extract local features
2. Learn “visual vocabulary”
3. Quantize local features using visual vocabulary
4. Represent images by frequencies of “visual words”

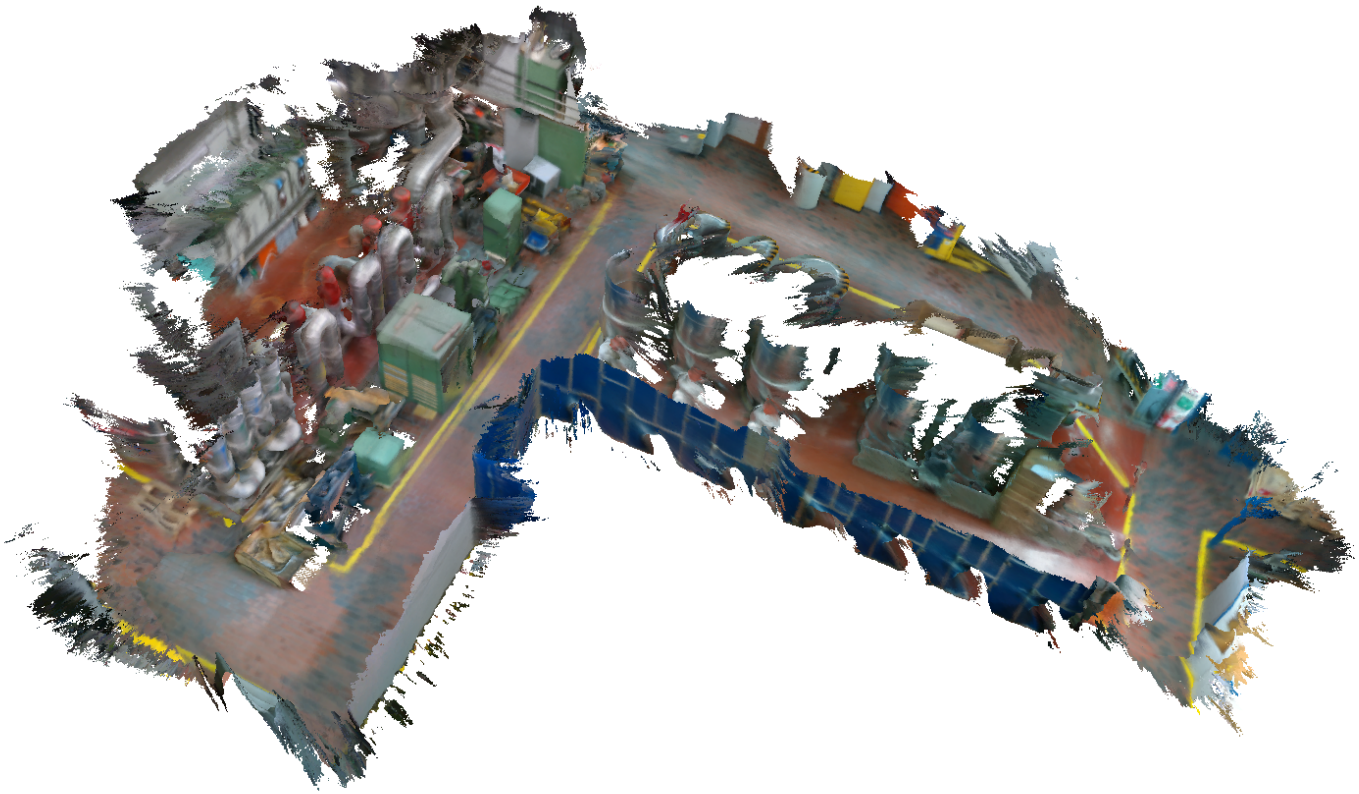


Slide credit: Svetlana Lazebnik

- DBoW2/3
  - Paper: http://doriangalvez.com/papers/GalvezTRO12.pdf
  - Code: https://github.com/rmsalinas/DBow3
- HBST: A Hamming Distance embedding Binary Search Tree
  - Paper: https://arxiv.org/abs/1802.09261
  - Code: https://gitlab.com/srrg-software/srrg_hbst

- Applications:
  - SfM: Speed up pairwise matching
  - SLAM: re-localization, loop-closure

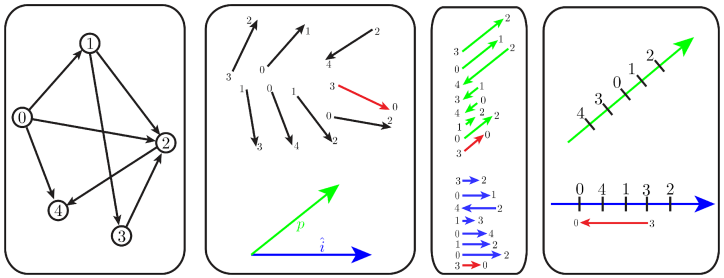
Dense 3D reconstruction for SfM

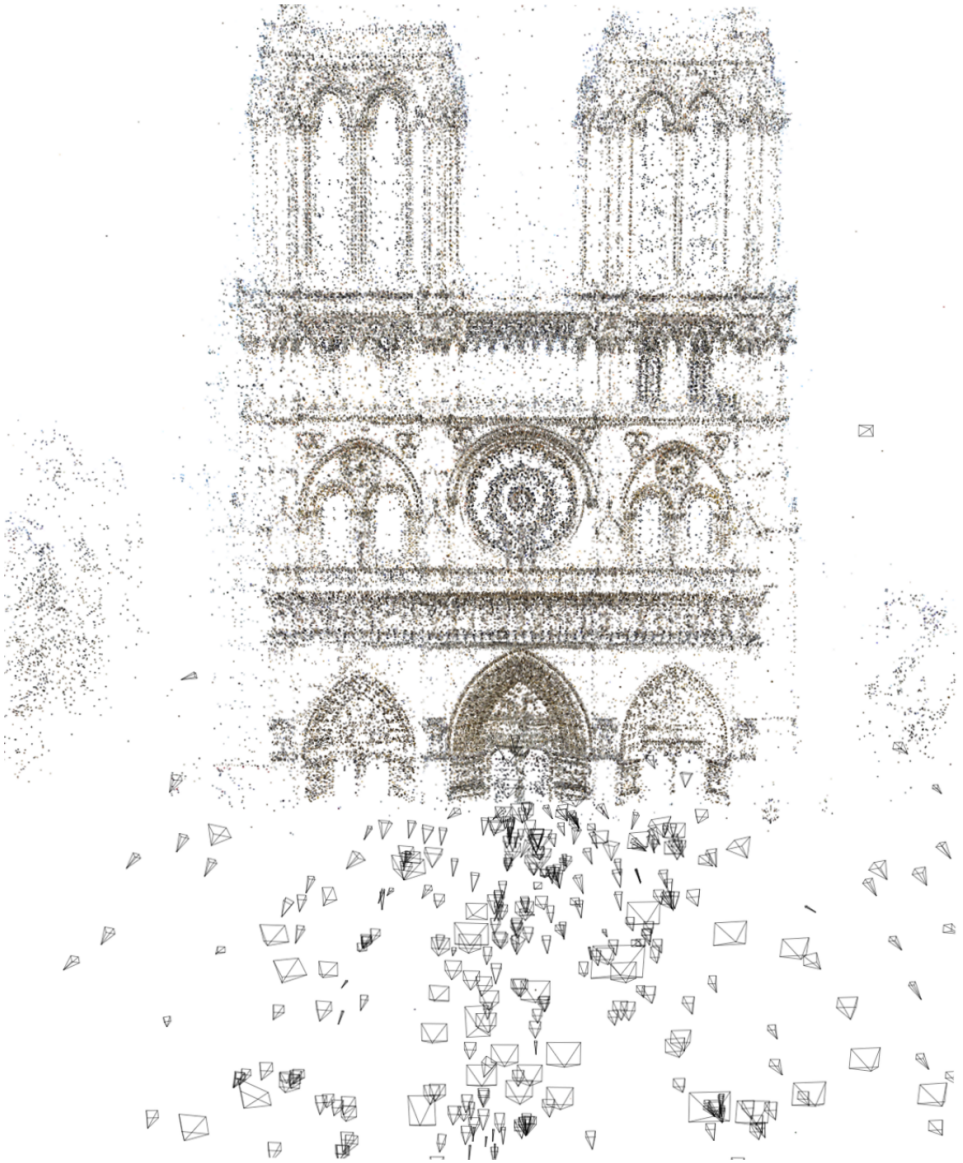


- PSMNet
  - https://arxiv.org/abs/1803.08669
  - https://github.com/JiaRenChang/PSMNet
- Depth images Fusion
  - Voxblox: https://github.com/ethz-asl/voxblox

Global SfM with Motion Averaging

- Goal: Implement global SfM pipeline using Motion Averaging (as opposed to the incremental pipeline from sheet 4)
- Approach:
  - Estimate relative rotation between pairs of cameras
  - Solve for global camera orientations
  - Given the global orientations, estimate global translations
  - Triangulate structure





- Chatterjee, Avishek, and Venu Madhav Govindu. "Efficient and robust large-scale rotation averaging." Proceedings of the IEEE International Conference on Computer Vision. 2013.